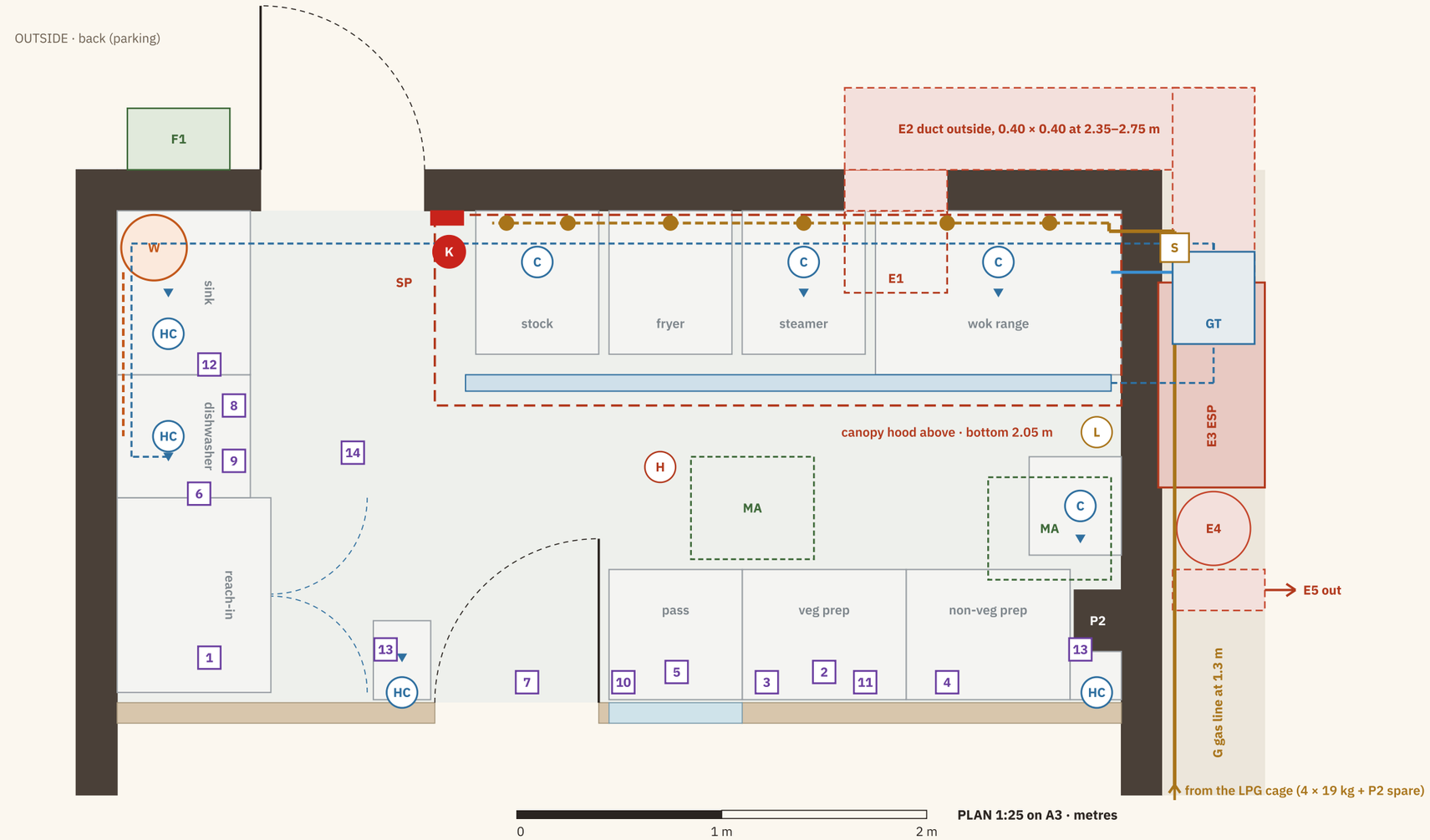
Kitchen services and exhaust

For kitchen equipment, exhaust, gas, plumbing and electrical vendors · plan at 1:25 and elevation at 1:40 on A3



Services list

TAG	ITEM	SIZE / NOTE
E1	Duct through the back wall	0.40 × 0.40 at 2.35–2.75 m
E2	Duct along the wall, round the corner	0.40 × 0.40, outside
E3	ESP + carbon odour filter	0.50 × 1.00 × 0.90, 2.05–2.95 m
E4	Inline exhaust fan	Ø 0.44 · on backup
E5	Side discharge, mesh + rain cowl	0.40 × 0.40 at ~2.6 m
F1	Fresh-air intake fan, back wall	0.50 × 0.48 at 3.3 m
MA	Make-up air diffusers ×2, ceiling	0.60 × 0.50 each · ~2.4 m/s
CC	Cooling coil in the make-up duct (P2)	1.5 TR · 3rd condenser on the wall
G	LPG line, copper, from the cage	at 1.3 m, in at the back corner
S	Solenoid valve on the entry, outside	interlocked: fan + suppression
SP	Hood suppression pull + gas cut-off	at the rear exit, 1.3–1.5 m
GT	Grease trap, outside back corner	0.45 × 0.45
W	Storage water heater, 3 kW	35 L · left wall over the sink
PF	Pot filler, cold, swing arm	over the stock range
WC	Water cabinet: pump, filter, softener (P2)	outside, 1.2–2.0 m · on backup
SU	Water sump, buried (P2)	1,000 L · 1.3 × 1.0 × 0.8 · garden
1	Reach-in fridge	1-ph socket · on backup
2	Under-counter freezer	1-ph socket · on backup
3	Two rice cookers	twin 1-ph socket
4	Chilled mise-en-place rail	1-ph socket · on backup
5	Heat lamps over the pass	1-ph, switched
6	Dishwasher (P2)	1-ph or 3-ph: confirm model
7	Insect killer	1-ph
8	Hood-light switch	left wall, by panel 9
9	Exhaust fan + ESP control panel	left wall over the DW · on backup
10	Pass under-counter fridge (P2)	1-ph socket · on backup
11	Mixer-grinder (P2)	1-ph socket, 16 A
12	Water heater isolator	own 3 kW MCB
13	Hand-wash heaters ×2	1 kW each, 6 L
14	Goods hoist over the hatch (P2)	200 kg · 0.4 kW

EXHAUST + AIR

Allow about 4,000–5,000 m³/h for the whole hood (vendor to size); about 8 m/s in the 0.40 × 0.40 duct. ESP + carbon filter before the fan. No roof stack: apartment above, parking behind. Hood overhang 0.20 m at the open end, 0.15 m over the wok range. Make-up air about 80% of the exhaust through two low-velocity diffusers (~2.4 m/s), interlocked; the 1.5 TR cooling coil is P2.

GAS

4 × 19 kg LPG on the manifold (a 5th full spare in P2 = 95 kg) in a locked cage on the right strip. Typical load: wok 2 × 18, stock 2 × 8, fryer 18, steamer 11 kW = about 80 kW connected, 55 kW at once; 4 cylinders give about 60–75 kW in Goa: vendor to confirm the burner ratings, add a vaporiser if they are higher. A valve per appliance; solenoid shuts off with the fan or the suppression; leak detector low on the right wall.

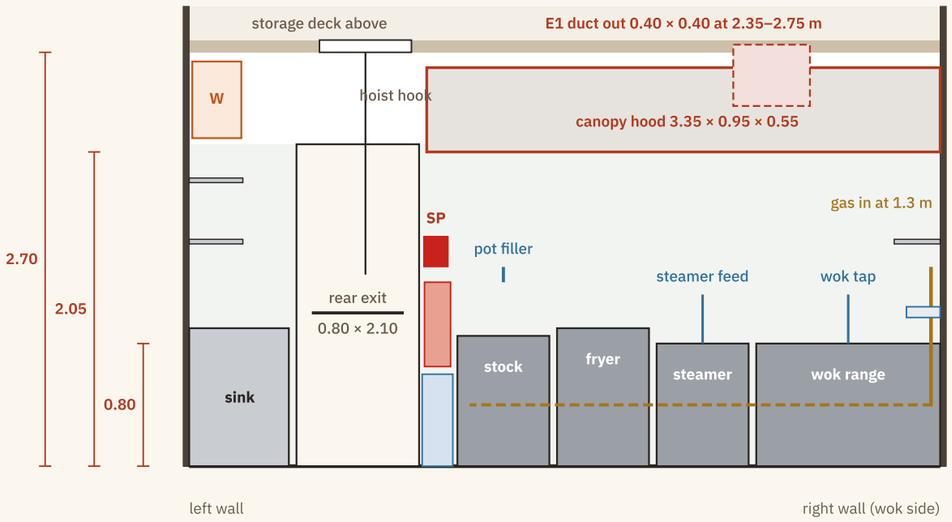
WATER + DRAINS

Cold: the existing supply in P1; P2 adds a 1,000 L buried sump → pump, filter, softener cabinet (on backup). Hot: 35 L heater for the sink and pre-rinse, 6 L heaters under both hand-wash basins. All kitchen drains to the grease trap, then the grey-water line (never the biodigester). Dishwasher with its own booster in P2.

POWER

Shop supply: three-phase, about 15–20 kW. Backup 7.5 kVA covers the fridges, rails, fan, ESP, lights and the water pump. Not the ACs.

Hot line elevation · back wall seen from inside · 1:40 on A3



Legend

- canopy hood outline (above)
- exhaust duct, outside, at 2.35–2.75 m
- gas line, copper
- gas manifold at 0.40 m · dot = appliance valve
- S — solenoid shut-off valve
- L — gas leak detector
- HC — water point: hot + cold (C = cold only)
- hot water from the 35 L heater (W)
- cold water main in (WM)
- drain point
- drain line to the grease trap
- 1 — power point, see the services list
- make-up air diffusers in the ceiling (MA, 2)
- H — heat detector
- SP suppression pull + gas cut-off · K = Class K extinguisher